# Olivia Chiarini, she/they

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## Education

**Drexel University** Jan. 2025 – Present
B.S. in Physics GPA: 3.67
Transferred from **University of Maryland, Baltimore County**, Aug. 2022 – December 2024

## Research Experience

**Data Analysis and Run Control for the Scintillating Bubble Chamber** Mar. 2026 – Present
*Drexel University and Fermilab; PI: Russell Neilson, Ph.D.*

- Worked in person at Fermilab on the SBC, including maintenance and data collection runs
- Updated documentation for access and analysis of SBC events
- Created reconstructions of events with bubble positions and size using multiple detector types including silicon photomultipliers and piezoelectrics
- Developed algorithm for bubble multiplicity calculation and visualization
- Led handscanning campaign for tens of thousands of nucleation events

**Astronomical Data Reduction Pipeline** Aug. 2023 – Jan. 2025
*University of Maryland, Baltimore County (UMBC); PI: Eileen Meyer, Ph.D.*

- Developed an astronomical data reduction pipeline
- Benchmarked said pipeline against the Landolt Standard Stars as a part of a UMBC Undergraduate Research Award
- Developed data analysis pipeline to determine the FWHM of astronomical sources and aid in the characterization of the atmosphere around UMBC

**Execution of Density Functional Theory Workflows on HPC Cluster** Jun. 2024 – Aug. 2024
*SCIART Summer Program, UMBC; PI: Joseph Bennet, Ph.D.*

- Used UMBC's High Performance Computing Cluster for Density Functional Theory research as part of Andrew W. Mellon Foundation's SCIART Internship
- Analyzed the usage of Titanium Carbide MXenes as films to protect against harmful adsorptions
- Presented poster and abstract at UMBC's Summer of Undergraduate Research Festival

**Simulating the AXIS Space Telescope Observation Scheduler** Mar. 2023 – Sep. 2023
*UMBC; PI: Eileen Meyer, Ph.D.*

- Optimized computer simulations for satellite scheduling
- Developed procedures for scheduled observations and recovery from time-sensitive interruptions

## Professional Experience

**Teaching Assistant** May 2025 – Present
*Drexel University*

- Graded homework, labs, exams, and quizzes for introductory physics classes
- Supported and ran lab classes for introductory physics classes in mechanics, electricity & magnetism, thermodynamics, and optics
- Supported and ran discussion sections for small group problem solving
- Wrote, proctored, and distributed exams for large groups of physics students

**Senior Telescope Operator and Research Analyst** Nov. 2023 – Dec. 2024
*University of Maryland, Baltimore County*

- Created the current data reduction pipeline
- Delivered and discussed physics demonstrations to large audiences
- Wrote and delivered short talks on topics in astronomy to large audiences
- Developed astronomical imaging pipelines for processing astronomical imaging using Python, acted as software maintainer in GitHub by fixing bugs and adding requested features

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**i3b's The Pride Center Student Supervisor** Aug. 2022 – May. 2024

*University of Maryland, Baltimore County*

- Planned and hosted social and educational events/trainings for over 500 unique students and staff.
- Helped organize and process data for student engagement in MS Excel.
- Supervised, managed, and trained student staff interns throughout i3b on event planning, conflict resolution, and communication.
- Organized regular meetings with professional and student staff, planning agendas and running said meetings as necessary.
- Worked to engage directly with undergraduates, graduate students, faculty, and staff.

## Technical and Academic Skills

- **Physics:** Observational Astronomy, Galactic Astrophysics, Cosmology, General Relativity, Neutrino Physics, Particle Physics, Statistical Mechanics, Electricity & Magnetism, Quantum Mechanics, Classical Mechanics, Optics w/Lab, Electronics w/Lab
- **Math:** Linear Algebra, Differential Equations, Multivariable Calculus, Differential Geometry, Tensor Calculus
- **Computing:** Comfortable with a variety of operating systems: Windows, macOS, Ubuntu, Pop_OS, Arch. Experience programming in Python, C, C++, C#, Java, and Bash. Algorithm optimization with IBM CPLEX Engine. Proficient in MS Office Suite, Google ecosystem, and LaTeX.
- **Logistics & Planning**: Comfortable working with a variety of institutional partners to plan and execute large scale events (300–350 people).

## Presentations and Publications

- Chiarini, O. (2026, November 13–15). Scintillating Bubble Chamber Liquid Argon 10 kg (SBC-LAr10) Detector Analysis [Conference presentation]. APS Mid-Atlantic Section Annual Meeting 2026, Baltimore, MD, United States
- Chiarini, O., Doherty, C., Bennett, J., Rosenzweig, Z. 2-DM (2024, August 7th) MXenes as Chemical Sensors in the art museum environment [Conference presentation]. UMBC's Summer Undergraduate Research Fest, Baltimore MD, United States

## Leadership, Accolades, & Volunteer Experience

- The Susan and Donald Larson Endowed Scholarship, 2026
- Barry Goldwater Scholarship and Excellence in Education Foundation Finalist, 2026
- Drexel Kaczmarczik Day Volunteer 2025, 2026
- Drexel Dean's List Winter 2025, Spring 2025, Summer 2025, Fall 2025
- A. J. Drexel Scholarship 2025, 2026
- UMBC Undergraduate Research Award, 2024
- Vice President, UMBC Astronomy Club 2024-2025
- Delivered outreach at Astronomy Festival on the National Mall June 22, 2024
- Andrew W. Mellon Foundation SCIART Internship, Summer 2024
- Proctored 2024 Science Olympiad Division C Invitational Astronomy Exam Spring 2024
- Recipient of State Delegate Kathy Forbes Scholarship 2022, 2023, 2024
- Recipient of State Senator Chris West Scholarship 2022, 2023, 2024
- Recipient of George Washington Carver Center IT/IMP Scholarship 2022